Customer Shopping Behavior Analysis

Project Overview

This project analyzes customer shopping behavior using transactional Data

From 3,900 purchases across various product categories. The goal is to uncover insights into spending

patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

Dataset Summary

1. Rows: 3,900

2. Columns: 18

3. Key Features:

- Customer demographics (age, gender, location, subsription_status)
- Purchase details (item_purcahsed, category, purchase_amount, season, size, color)
- Shopping behaviors (discount_applied, promo_code_used, previous_purchases, frequency_of_purchases, review_rating, shipping_type)

4. Missing Data: 37 values in Review Rating Column

```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating             37
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

Exploratory Data Analysis

1. Data preparation and cleaning:

- Data Loading: imported dataset using pandas

2. Initial Exploration: used df.info () to check structure and. describe () to summary statistics

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

3. Handling Missing Data: checked for nulls and imputed missing values in the review_rating column using the median ration of each product category

```
Customer ID               0
Age                       0
Gender                    0
Item Purchased            0
Category                  0
Purchase Amount (USD)     0
Location                  0
Size                      0
Color                     0
Season                    0
Review Rating             0
Subscription Status       0
Shipping Type             0
Discount Applied          0
Promo Code Used           0
Previous Purchases        0
Payment Method            0
Frequency of Purchases    0
dtype: int64
```

4. Column Standardization: Renamed columns to snake case for better readability and documentation

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

5. Feature Engineering:

- Created age_group column by binning customer ages

	age	age_group
0	55	Middle Age
1	19	Young Adult
2	50	Middle Age
3	21	Young Adult
4	45	Middle Age
5	46	Middle Age
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle Age

- Created purchase_frequency_days column from purchase data

	frequency_of_purchases	purchase_frequency_days
0	Fortnightly	14.0
1	Fortnightly	14.0
2	Weekly	7.0
3	Weekly	7.0
4	Annually	365.0
5	Weekly	7.0
6	Quarterly	90.0
7	Weekly	7.0
8	Annually	365.0
9	Quarterly	90.0

6. *Data Consistency Check:* Verified if discount_applied and promo_code_used were redundant, and dropped promo_code_used

7. *Database Integration:* Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis

```
Data has been successfully loaded into the 'customer' table in the PostgreSQL database.
```

Data Analysis using SQL

- Revenues by gender

A-Z gender	123 revenue
Female	75,191
Male	157,890

- High-spending discount users

	123 customer_id	123 purchase_amount
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88

- Top 5 products by ratings

A-Z item_purchased	123 average_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

- Average purchase amount by standard vs. express shipping

A-Z shipping_type	123 avg_purchase_amount
Standard	58.46
Express	60.48

- Subscribers vs. non-subscribers

subscription_status	total_customers	avg_spend	total_revenue
Yes	1,053	59.49	62,645
No	2,847	59.87	170,436

- Top 5 purchased products with discount discount_applied

item_purchased	discount_rate
Hat	50
Sneakers	49
Coat	49
Sweater	48
Pants	47

- Segments of customers

customer_segment	Number of Customers
Loyal	3,116
New	83
Returning	701

- Top 3 purchased products in each category

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

- Repeat buyers & subscription

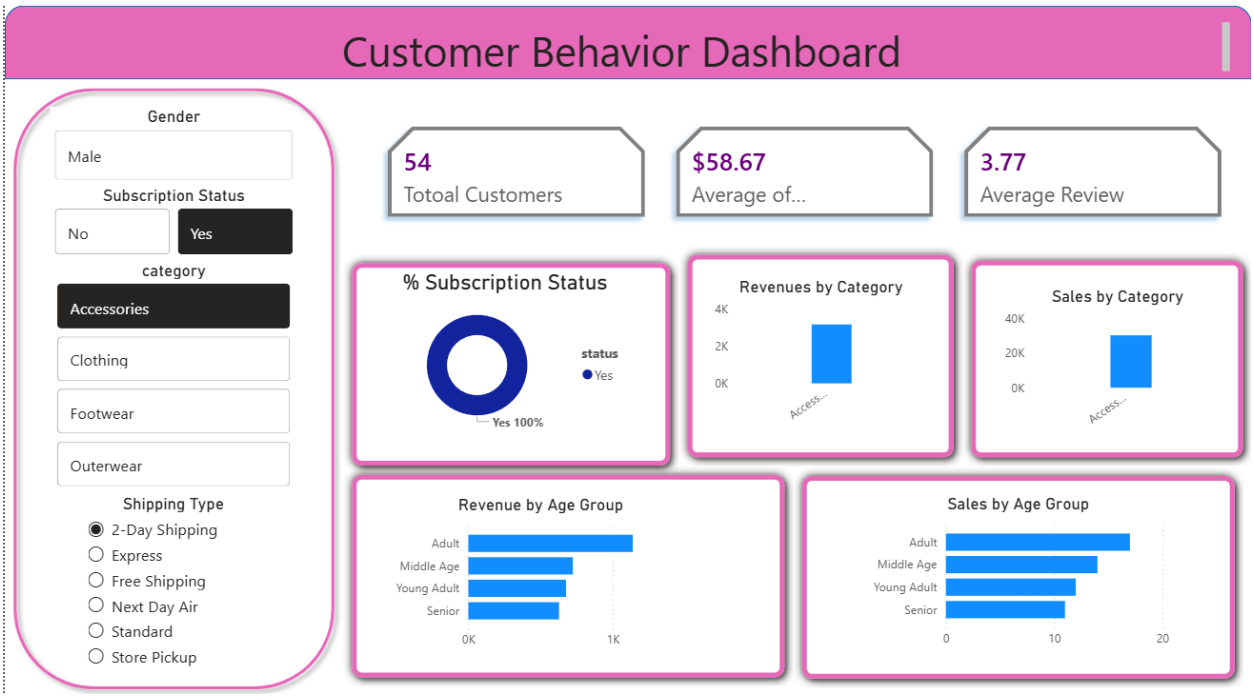
subscription_status	repeat_buyers
No	2,518
Yes	958

- Revenue by Age Group

age_group	total_revenue
Young Adult	62,143
Middle Age	59,197
Adult	55,978
Senior	55,763

Dashboard in Power BI

Interactive dashboard was developed in Power BI to present insights visually



Business Recommendations

* Boost subscriptions: promote exclusive benefits to subscribers

* Customer loyalty programs: reward-move repeat buyers to into 'loyal' segment

* Review discount policy: balance sales boosts with margin control

* Product positioning: highlight top-rated and best-selling products in campaigns

* Target Marketing: focus on high-revenue age groups and express-shipping users